CSA0614-TOPIC 6-DAA

Y Greeshma Reddy

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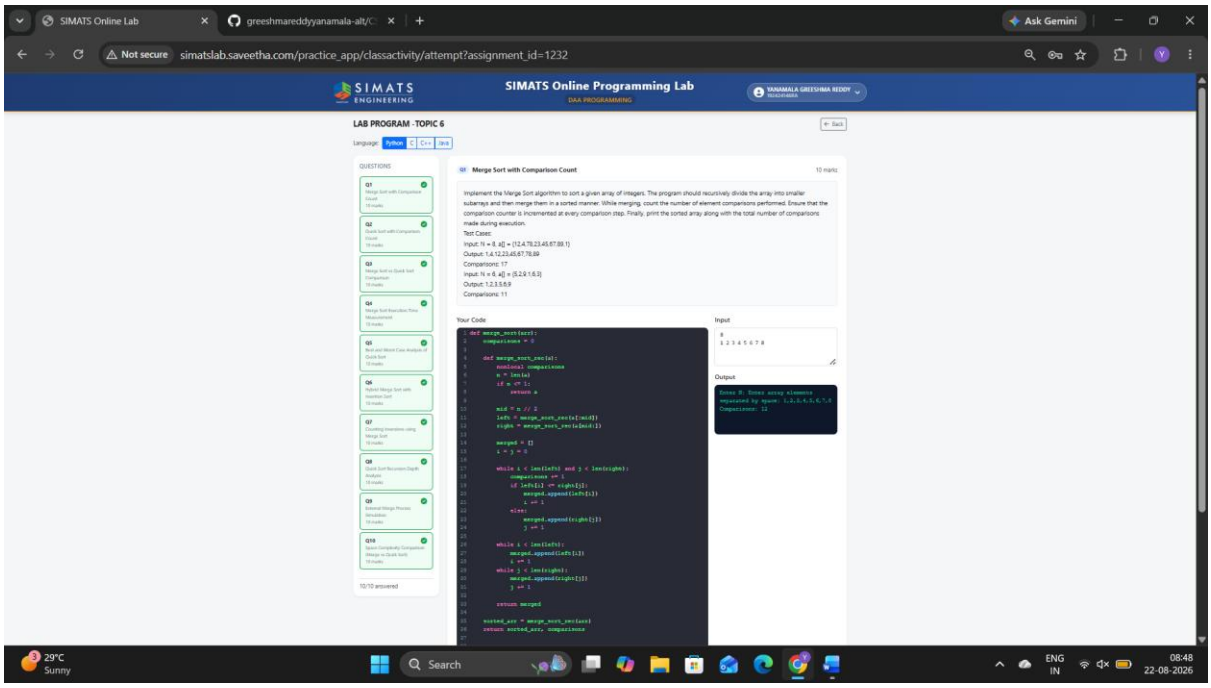

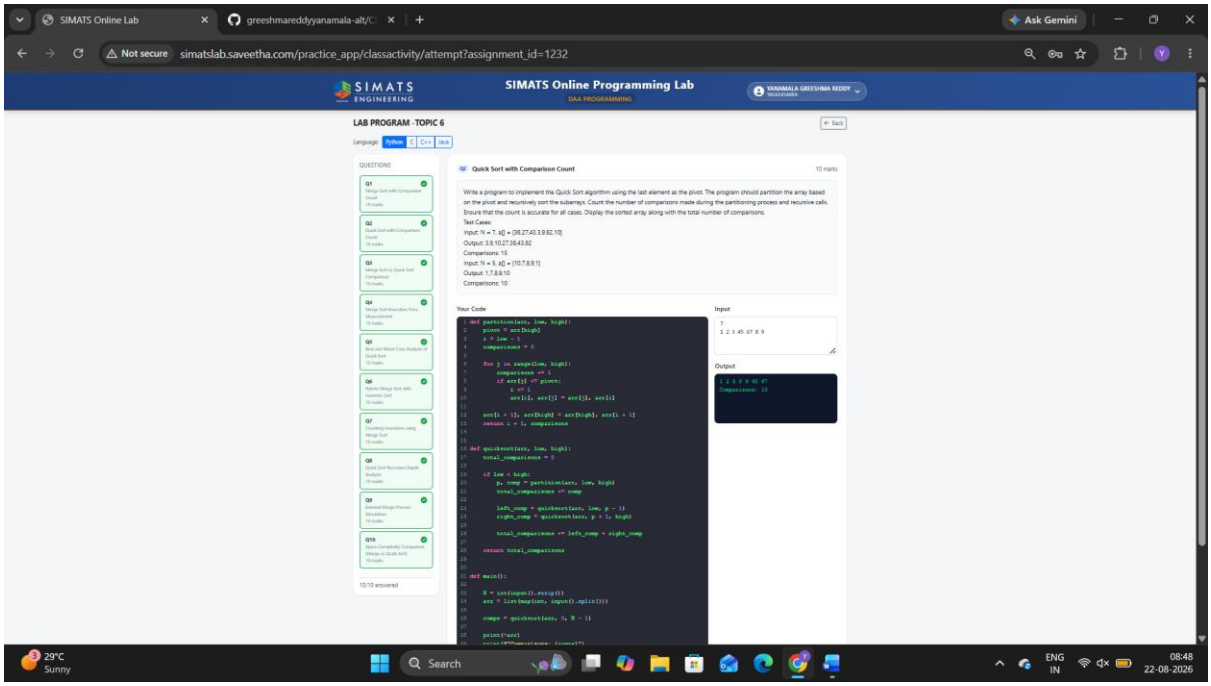

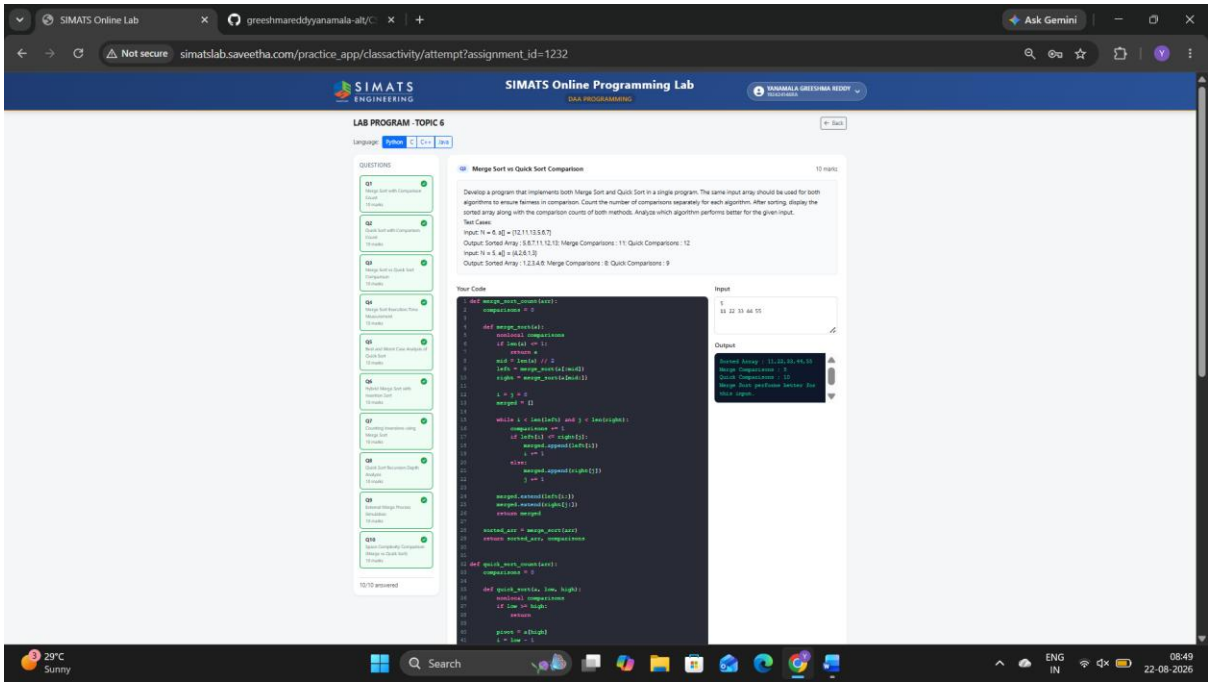
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Back
Language: Python C C++ Java
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Merge Sort vs Quick Sort Comparison
10 marks
Develop a program that implements both Merge Sort and Quick Sort in a single program. The same input array should be used for both algorithms to ensure fairness in comparison. Count the number of comparisons separately for each algorithm. After sorting, display the sorted array along with the comparison counts of both methods. Analyze which algorithm performs better for the given input.
Test Cases:
Input: N = 6, a[] = {12,11,13,5,6,7}
Output: Sorted Array : 5,6,7,11,12,13: Merge Comparisons : 11: Quick Comparisons : 12
Input: N = 5, a[] = {4,2,6,1,3}
Output: Sorted Array : 1,2,3,4,6: Merge Comparisons : 8: Quick Comparisons : 9
Your Code
Input
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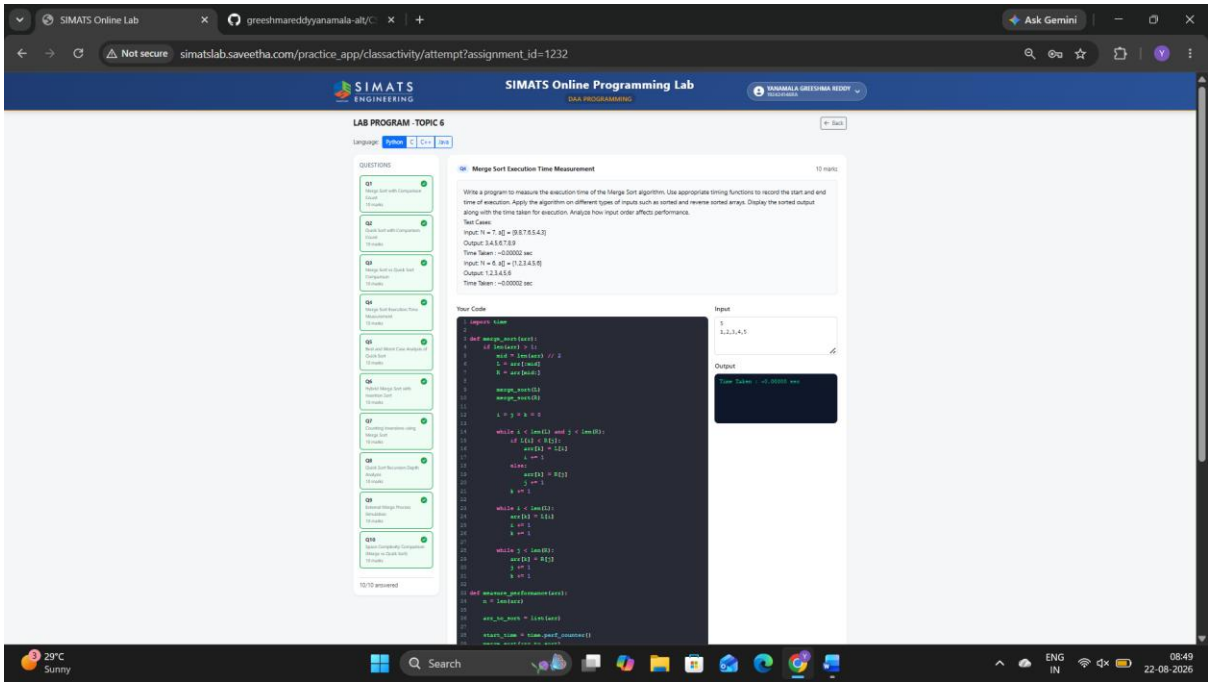
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Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Merge Sort Execution Time Measurement
10 marks
Write a program to measure the execution time of the Merge Sort algorithm. Use appropriate timing functions to record the start and end time of execution. Apply the algorithm on different types of inputs such as sorted and reverse sorted arrays. Display the sorted output along with the time taken for execution. Analyze how input order affects performance.
Test Cases:
Input: N = 7, a[] = {9,8,7,6,5,4,3}
Output: 3,4,5,6,7,8,9
Time Taken : ~0.00002 sec
Input: N = 6, a[] = {1,2,3,4,5,6}
Output: 1,2,3,4,5,6
Time Taken : ~0.00002 sec
Your Code
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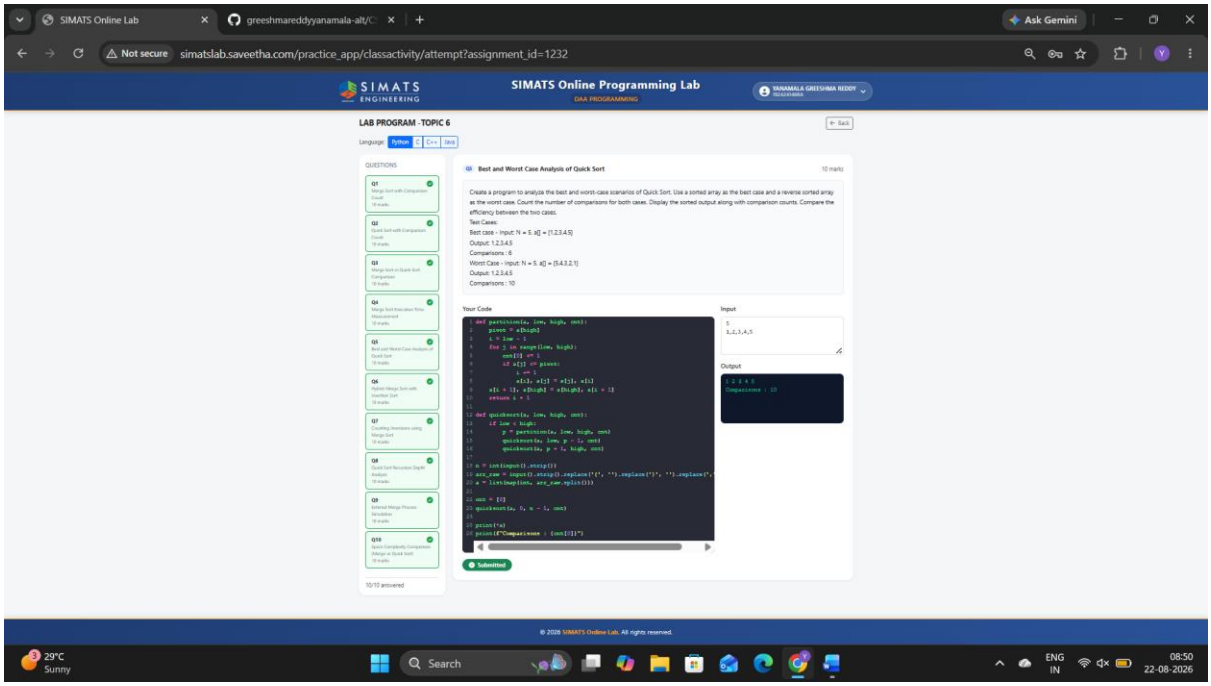
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QUESTIONS
Best and Worst Case Analysis of Quick Sort
10 marks
Create a program to analyze the best and worst-case scenarios of Quick Sort. Use a sorted array as the best case and a reverse sorted array as the worst case. Count the number of comparisons for both cases. Display the sorted output along with comparison counts. Compare the efficiency between the two cases.
Test Cases:
Best case - Input: N = 5, a[] = [1,2,3,4,5]
Output: 1,2,3,4,5
Comparisons : 6
Worst Case - Input: N = 5, a[] = [5,4,3,2,1]
Output: 1,2,3,4,5
Comparisons : 10
Your Code
Input
Output
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10/10 answered
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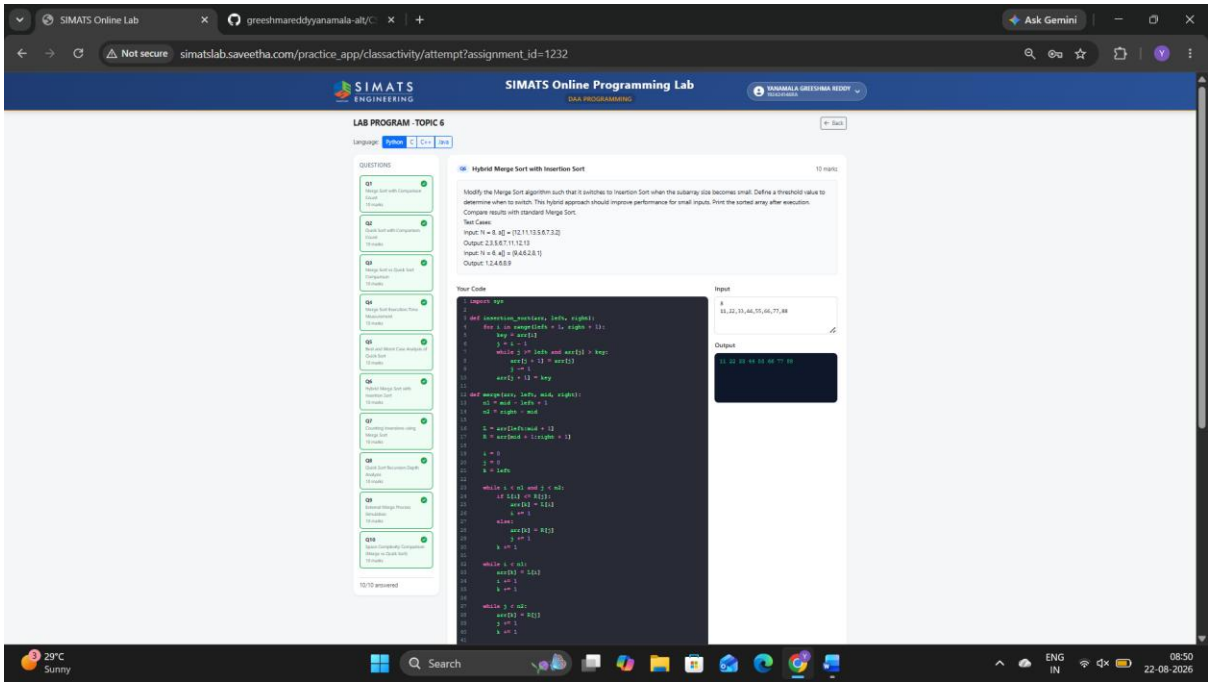
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QUESTIONS
Hybrid Merge Sort with Insertion Sort
10 marks
Modify the Merge Sort algorithm such that it switches to Insertion Sort when the subarray size becomes small. Define a threshold value to determine when to switch. This hybrid approach should improve performance for small inputs. Print the sorted array after execution. Compare results with standard Merge Sort.
Test Cases:
Input: N = 8, a[] = [12,11,13,5,6,7,3,2]
Output: 2,3,5,6,7,11,12,13
Input: N = 6, a[] = [9,4,6,2,8,1]
Output: 1,2,4,6,8,9
Your Code
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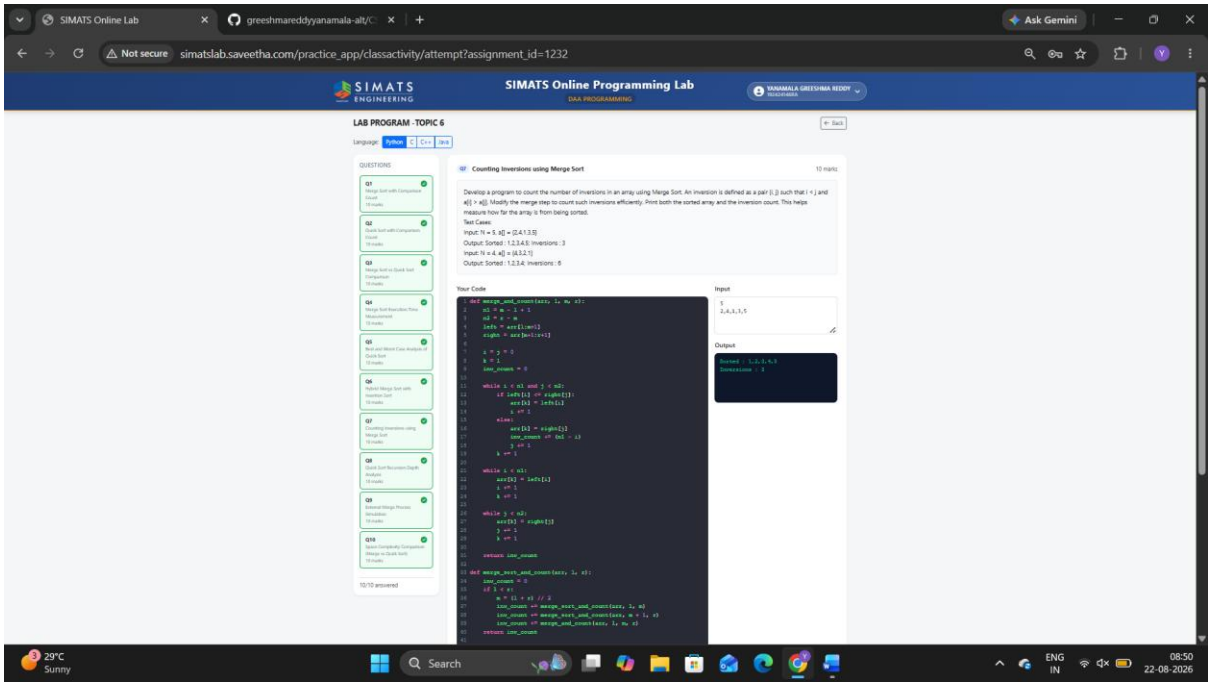
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Counting Inversions using Merge Sort
10 marks
Develop a program to count the number of inversions in an array using Merge Sort. An inversion is defined as a pair (i, j) such that i < j and a[i] > a[j]. Modify the merge step to count such inversions efficiently. Print both the sorted array and the inversion count. This helps measure how far the array is from being sorted.
Test Cases:
Input: N = 5, a[] = [2,4,1,3,5]
Output: Sorted : 1,2,3,4,5 Inversions : 3
Input: N = 4, a[] = [4,3,2,1]
Output: Sorted : 1,2,3,4 Inversions : 6
Your Code
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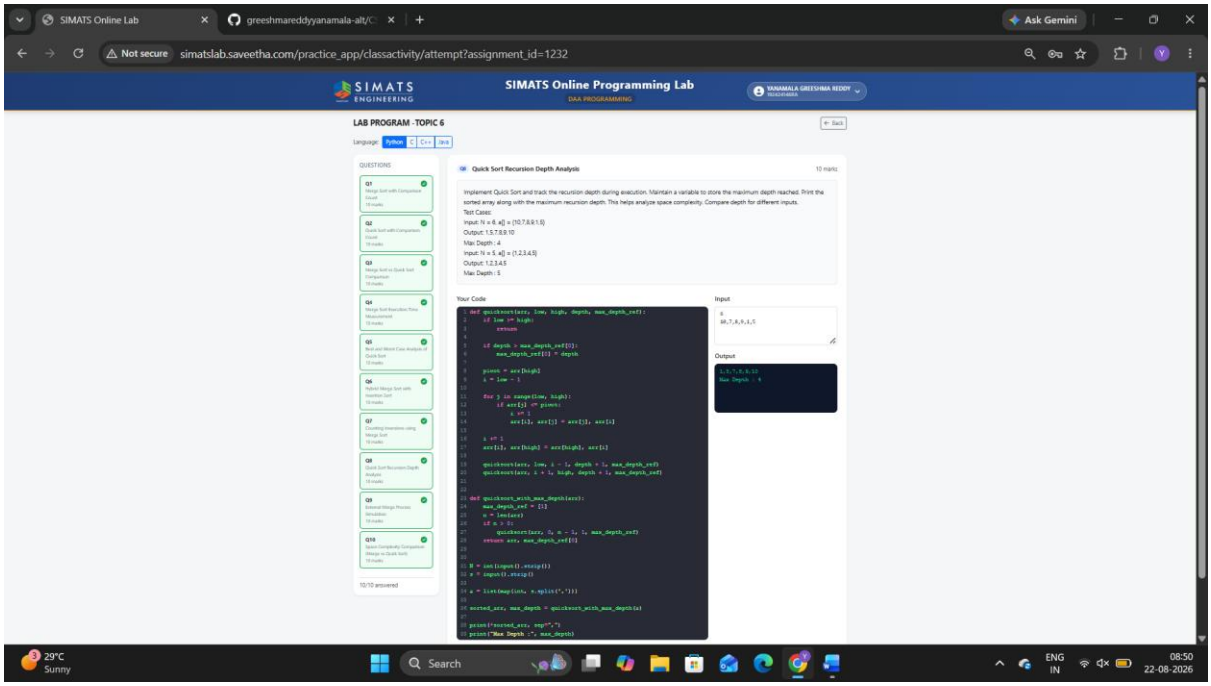
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Quick Sort Recursion Depth Analysis
10 marks
Implement Quick Sort and track the recursion depth during execution. Maintain a variable to store the maximum depth reached. Print the sorted array along with the maximum recursion depth. This helps analyze space complexity. Compare depth for different inputs.
Test Cases:
Input: N = 6, a[] = [10,7,8,9,1,5]
Output: 1,5,7,8,9,10
Max Depth : 4
Input: N = 5, a[] = [1,2,3,4,5]
Output: 1,2,3,4,5
Max Depth : 5
Your Code
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Output
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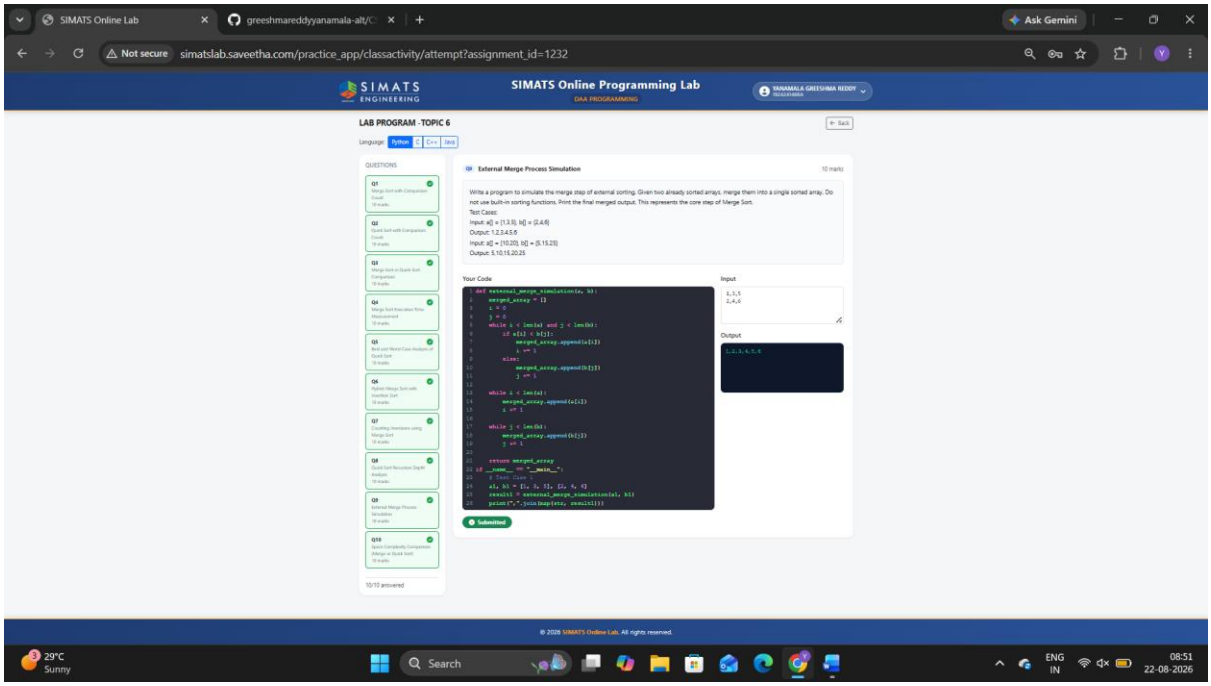
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QUESTIONS
External Merge Process Simulation
10 marks
Write a program to simulate the merge step of external sorting. Given two already sorted arrays, merge them into a single sorted array. Do not use built-in sorting functions. Print the final merged output. This represents the core step of Merge Sort.
Test Cases:
Input: a[] = [1,3,5], b[] = [2,4,6]
Output: 1,2,3,4,5,6
Input: a[] = [10,20], b[] = [5,15,25]
Output: 5,10,15,20,25
Your Code
Input
1,3,5
2,4,6
Output
1,2,3,4,5,6
Submitted
10/10 answered
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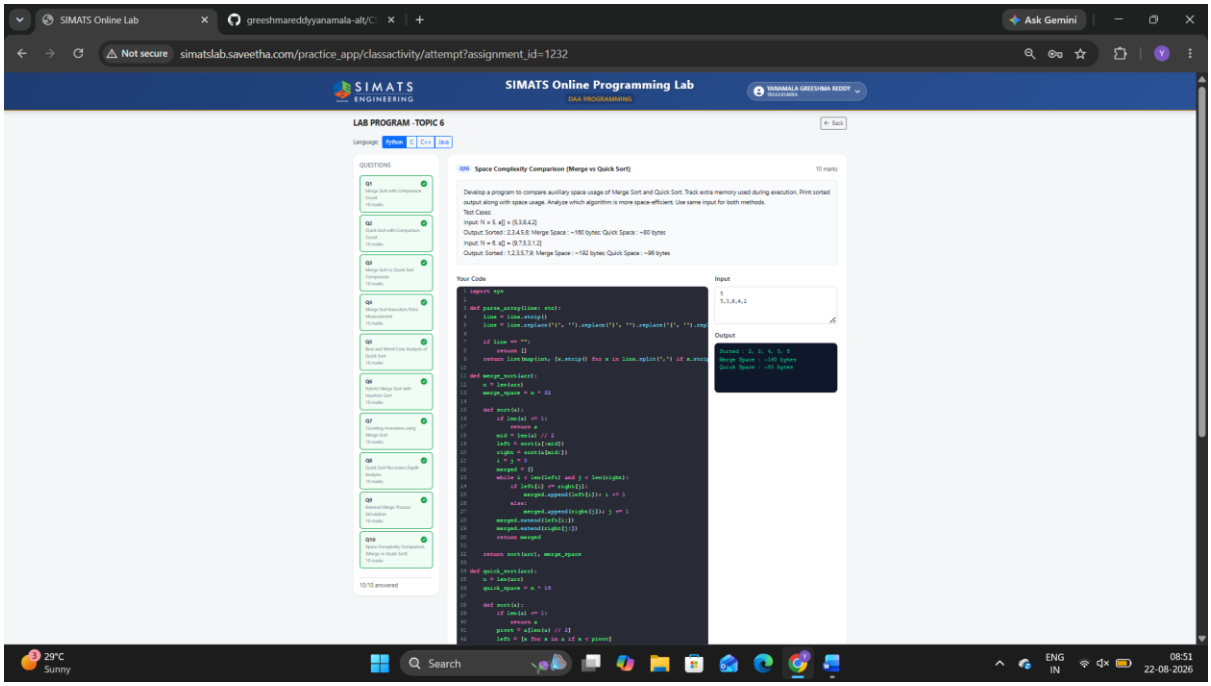
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Language: Python C C++ Java
QUESTIONS
Space Complexity Comparison (Merge vs Quick Sort)
10 marks
Develop a program to compare auxiliary space usage of Merge Sort and Quick Sort. Track extra memory used during execution. Print sorted output along with space usage. Analyze which algorithm is more space-efficient. Use same input for both methods.
Test Cases:
Input: N = 5, a[] = [5,3,8,4,2]
Output: Sorted : 2,3,4,5,8: Merge Space : ~160 bytes: Quick Space : ~80 bytes
Input: N = 6, a[] = [9,7,6,5,1,2]
Output: Sorted : 1,2,3,5,7,9: Merge Space : ~192 bytes: Quick Space : ~96 bytes
Your Code
Input
5
5,3,8,4,2
Output
Sorted : 2, 3, 4, 5, 8
Merge Space : ~160 bytes
Quick Space : ~80 bytes
10/10 answered
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